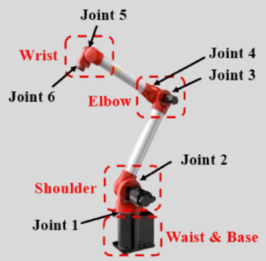
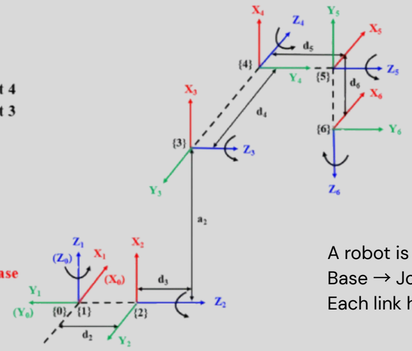
FORWARD KINEMATICS





1.DEFINITION & PURPOSE

Forward Kinematics is the process of finding the position and orientation of the robot's end-effector given:

Joint variables (angles or displacements)

Link parameters (lengths, offsets)

In simple terms: We know the joints → we calculate where the robot hand is.

Input vs Output

Input:

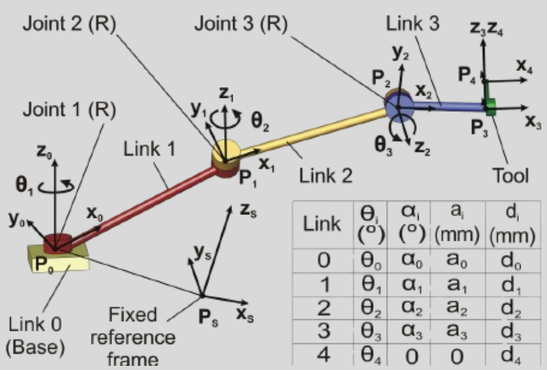
Joint angles (θ_1, θ_2, ...)

Link lengths

Output:

Position (x, y, z)

Orientation (rotation matrix)



Link	θ_i (°)	α_i (°)	a_i (mm)	d_i (mm)
0	θ_0	α_0	a_0	d_0
1	θ_1	α_1	a_1	d_1
2	θ_2	α_2	a_2	d_2
3	θ_3	α_3	a_3	d_3
4	θ_4	0	0	d_4

2. Robot Structure & Coordinate Frames

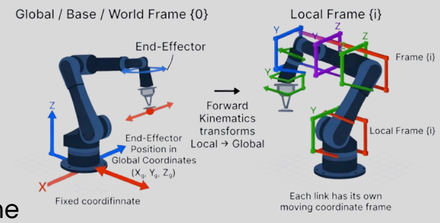
Robot as a Chain

A robot is a set of connected links:

Base → Joint → Link → Joint → Link → End-effector

Each link has its own coordinate frame.

Global Coordinates vs Local Coordinates in Robotics



Coordinate Frames

Each frame defines:

X-axis

Y-axis

Z-axis

Every link has its own local frame.

Global vs Local Frame (in Robotics)

In robotics, we describe the position and motion of every part of the robot using coordinate frames. The two most important ones are:

Global Frame (World/Base Frame)

Local Frame (Link Frame)

1. Global Frame (World / Base Frame)

The global frame is the fixed reference coordinate system of the entire robot.

It is attached to the robot base or the environment

It does NOT move during robot motion

it is important?

Because we need a universal reference to describe where everything is.

Without it, we cannot say "where the robot hand is in space

2. Local Frame (Link / Joint Frame)

A local frame is a coordinate system attached to each robot link or joint.

It moves together with the link

Every link has its own frame

Moves when the robot moves

Describes geometry of that specific link

Used to define joint motion

3. Relationship Between Global and Local Frames

This is the core idea of Forward Kinematics:

Each local frame is transformed step-by-step into the global frame.

We do this using transformation matrices.

Step-by-step idea

Start from the base (Global Frame)

Move to Frame 1 using T1

Move to Frame 2 using T2

Continue until end-effector

For 6 DOF

Global Frame (Frame 0)

Fixed at robot base

Reference for entire robot

Does NOT move

Everything is finally expressed relative to Frame 0

Local Frames (Frame 1 → Frame 6)

Each link has its own frame:

Moves with the link

Rotates when joint rotates

Used to describe local geometry

Forward Kinematics =

Convert every local frame step-by-step into the global frame.

1) Define the Base Frame (Global Frame)

This is your reference frame (Frame 0).

2) Assign a Frame to Each Link Using Denavit–Hartenberg (DH) parameters:

- θ (joint angle)
- d (offset)
- a (link length)
- α (twist)

3) Build Transformation Matrices

Each link has a transformation:

T1,T2,T3,...Tn

Each matrix represents:

Transformation from one local frame to the next

4) Multiply Them (Forward Kinematics)

T=T1·T2·T3…Tn

Important:

- Order matters
- This is based on local (moving) axes, not global

5) Final Result

T=[R P
O 1]

R (3×3)

→ Orientation of the end-effector (gripper)

P (3×1)

→ Position of the end-effector

3.Transformations in FK

A transformation describes:

Rotation (orientation change)

Translation (position change)

Homogeneous Transformation Matrix

$$T = \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix}$$

Where:

R = 3×3 rotation matrix

P = position vector (x, y, z)

Purpose

It allows us to:

Move between coordinate frames

Combine rotations + translations in one matrix

4. Denavit-Hartenberg (DH) Convention

To standardize how we describe robot links.

Instead of random math → we use 4 parameters per link.

- DH Parameters

1. θ (Theta)

Rotation around Z-axis

Joint angle (for revolute joints)

2. d

- Translation along Z-axis
- Joint offset (for prismatic joints)

3. a

- Link length
- Distance along X-axis

4. α (Alpha)

- Twist angle between Z-axes
- Rotation around X-axis

- Why DH is important?

Simplifies robot modeling

Makes FK systematic

5. DH Transformation Matrix

Each link has one transformation matrix.

Full Matrix Concept

It combines:

Rotation (θ, α)

Translation (a, d)

Final FK Formula

T=T1·T2·T3…Tn

Multiply all link transformations from base to end-effector.

6.Composition of Transformations

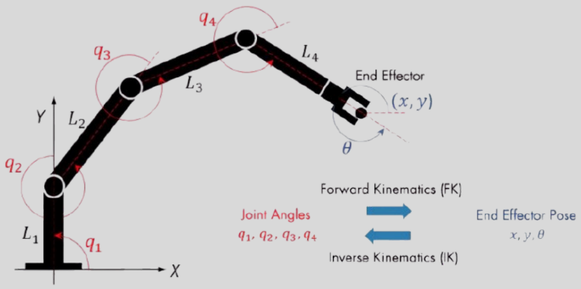
What it means

Robot motion is built step-by-step:

Each joint adds a transformation

Final pose is cumulative effect

INVERSE KINEMATICS



1) Definition of Inverse Kinematics

Inverse Kinematics (IK) is the process of determining the joint variables (angles or displacements) required to achieve a desired end-effector pose.

- Pose = Position (P) + Orientation (R)
- IK is the inverse problem of Forward Kinematics (FK)

Input vs Output

- Desired position:
- x,y,z
- Desired orientation:
- Rotation matrix (R)
- Euler angles or roll–pitch–yaw

Joint variables:

- θ1,θ2,...,θn . (revolute joints)
- d1,d2,...,dn (prismatic joints)

Key Idea:
FK: Joint variables → End-effector pose
IK: End-effector pose → Joint variables

2) Types of IK Solutions

Analytical (Closed-Form Solution)

- Exact mathematical solution
- Uses:
- Geometry
- Trigonometry (sin, cos, atan2)
- Very fast and accurate
- Used for:
- Simple manipulators (2DOF, 3DOF, some 6DOF with special structure)
- Limitation:
- Not always possible for complex robots

Numerical (Iterative Solution)

- Approximate solution
- Starts from an initial guess and iterates

Methods:

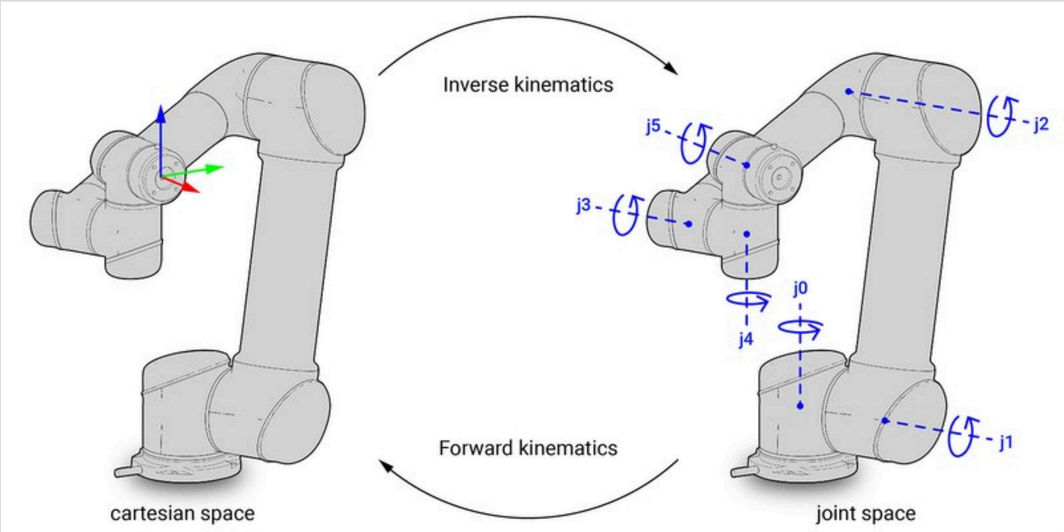
- Jacobian Inverse:
- θ˙=J^−1x˙
- Jacobian Transpose
- Gradient Descent

Advantages:

- Works for complex robots

Disadvantages:

- Slower
- May not converge
- Depends on initial guess



3) Workspace

Reachable Workspace

All points the robot can reach

Dexterous Workspace

Points where the robot can reach with any orientation

4) Singularities

Definition:
A singularity occurs when the robot loses one or more degrees of freedom (DOF)

Effects:

- Infinite number of solutions OR no solution
- Loss of control in certain directions
- Large joint velocities

5)Mathematical Tools Used in IK

Transformation Matrix

$$T = \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix}$$

Where:
R = 3×3 rotation matrix
P = position vector (x, y, z)

Trigonometry

Used in analytical IK:

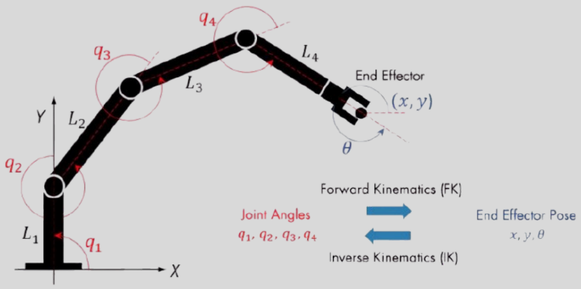
- sine, cosine
- inverse functions (atan2)

Jacobian Matrix

Relates joint velocities to end-effector velocities:

- x˙=J(θ)θ˙

INVERSE KINEMATICS



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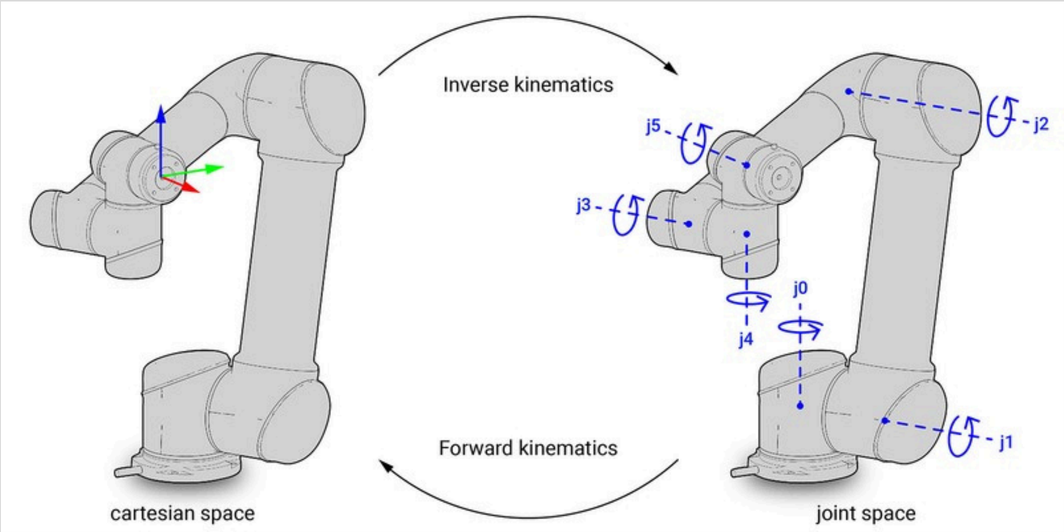

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